

Sjogren's Disease and more autoimmunity....

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LECOM

Case 1

- 54 yo lady complains of 1 yr of gritty sensation in eyes
- Mild photophobia at times with redness
- Dry mouth, multiple new cavities in past yr
- Arthralgias in small finger joints
- Diffuse skin itch
- Alkaline phosphatase 250

Henrik Samuel Conrad Sjögren
(July 23, 1899 – September 17, 1986)



Sjögren's disease – General points

- Idiopathic autoimmune dz affecting largely exocrine glands
 - environmental?
 - genetically susceptible (F:M is 9:1, HLA associations)
 - infectious (EBV, HTLV-1)?
- T cells activate B lymphocytes in glands, triggering autoimmune antibodies that attack **muscarinic receptors**
- Lymphocytic infiltration results in end organ damage

OLD Criteria for the classification of Sjögren's

1. Oral Symptoms (xerostomia, taste changes, fluid needs, caries)
2. Ocular Symptoms (Foreign body sensation, gritty, sandy)
3. Ocular Signs
 - Schirmer test ≤ 5 mm in 5 mins
 - Rose Bengal stain
4. Histopathologic evidence of salivary gland involvement (lymphocytic infiltrate)
5. Salivary gland involvement (scintigraphy, sialogram)
6. Autoantibodies
 - SSA/Ro, SSB/La, ANA, RF

(3-4 or > high sensitivity and specificity)

- Exclusions: lymphoma, sarcoid, **HIV** (DILS: diffuse infiltrative lymphocytosis syndrome), **HCV**, head and neck radiation, medications, GVHD

NEW Classification Criteria for Sjogren's syndrome

- April 2012: S. C. Shiboski. ACRheumatology classification criteria for Sjögren's syndrome: A data-driven, expert consensus approach in the Sjögren's International Collaborative Clinical Alliance Cohort. Arthritis Care & Research. April 2012

Table 7. Proposed classification criteria for SS*

The classification of SS, which applies to individuals with signs/symptoms that may be suggestive of SS, will be met in patients who have at least 2 of the following 3 objective features:

1. Positive serum anti-SSA/Ro and/or anti-SSB/La or (positive rheumatoid factor and ANA titer $\geq 1:320$)
2. Labial salivary gland biopsy exhibiting focal lymphocytic sialadenitis with a focus score ≥ 1 focus/4 mm²†
3. Keratoconjunctivitis sicca with ocular staining score ≥ 3 (assuming that individual is not currently using daily eye drops for glaucoma and has not had corneal surgery or cosmetic eyelid surgery in the last 5 years)‡

Prior diagnosis of any of the following conditions would exclude participation in SS studies or therapeutic trials because of overlapping clinical features or interference with criteria tests:

History of head and neck radiation treatment
Hepatitis C infection
Acquired immunodeficiency syndrome
Sarcoidosis
Amyloidosis
Graft versus host disease
IgG4-related disease

* We excluded participants with rheumatoid arthritis, systemic lupus erythematosus, scleroderma, or other connective tissue disease from these analyses since there were only 87 (6%) such participants. SS = Sjögren's syndrome; ANA = antinuclear antibody.

† Using histopathologic definitions and focus score assessment methods as previously described (15).

‡ Using ocular staining score as previously described (17).

Sicca syndrome clinical manifestations

- Keratoconjunctivitis sicca
 - Ocular dryness
 - Corneal injury
- Xerostomia
 - Oral dryness
 - Dysphagia
 - Dental caries
 - Thrush
- Nasal dryness/epistaxis
- Vaginal dryness
 - Dyspareunia
 - Candidiasis



Sjögren's: other clinical and associated features

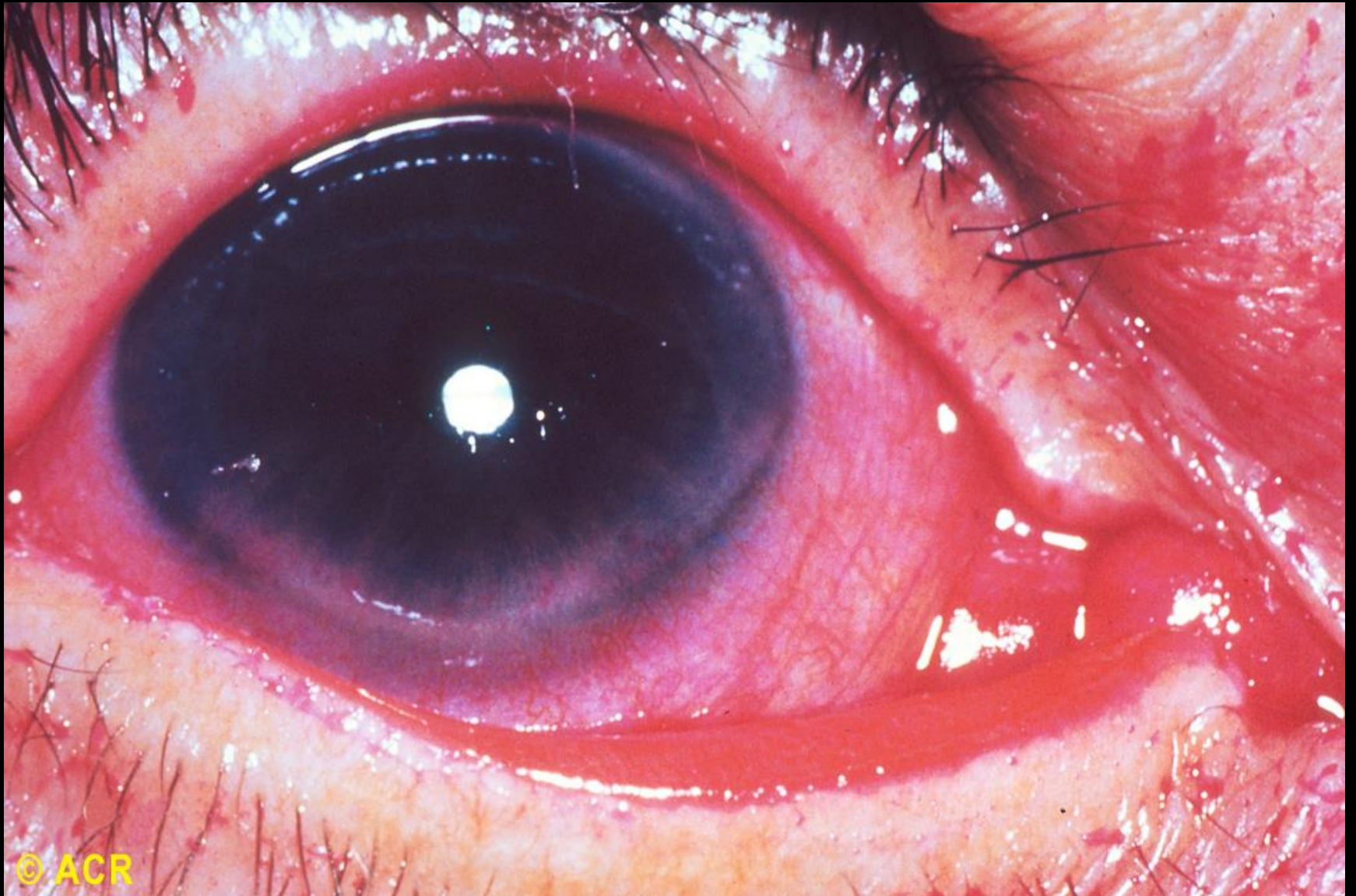
- Connective tissue diseases
 - SLE / RA
 - Systemic sclerosis
 - Vasculitis
 - leukocytoclastic
 - cryoglobulinemia
- Central, peripheral neurologic manifestations
- Nephropathy
 - Distal renal tubular acidosis
 - Interstitial nephritis
- Lymphoma
- Primary biliary cholangitis (anti-mitochondrial antibodies – AMA)
 - also F:M ratio of 9:1



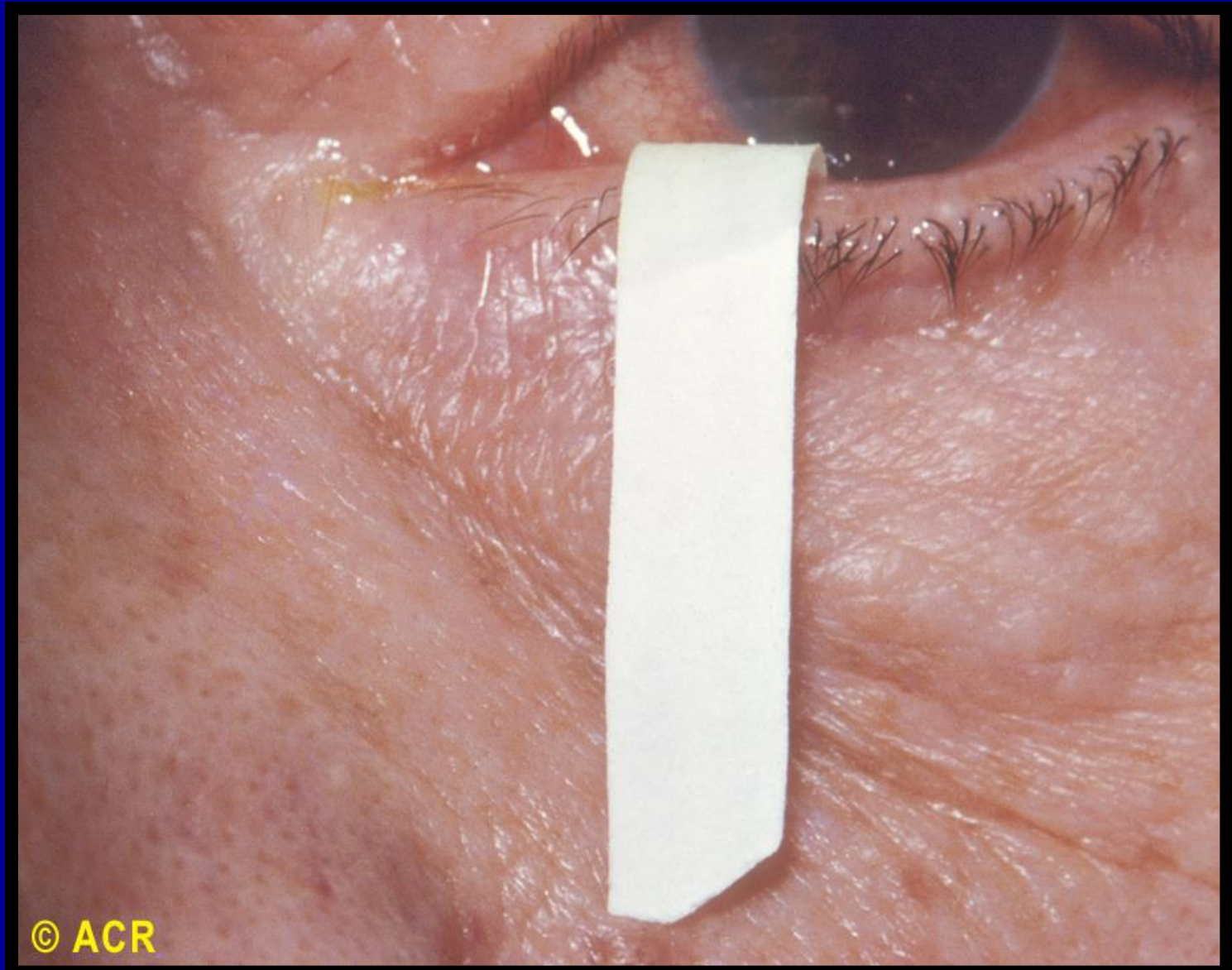
Sjögren's disease: parotid gland



Sjögren's: cornea (Rose Bengal stain)



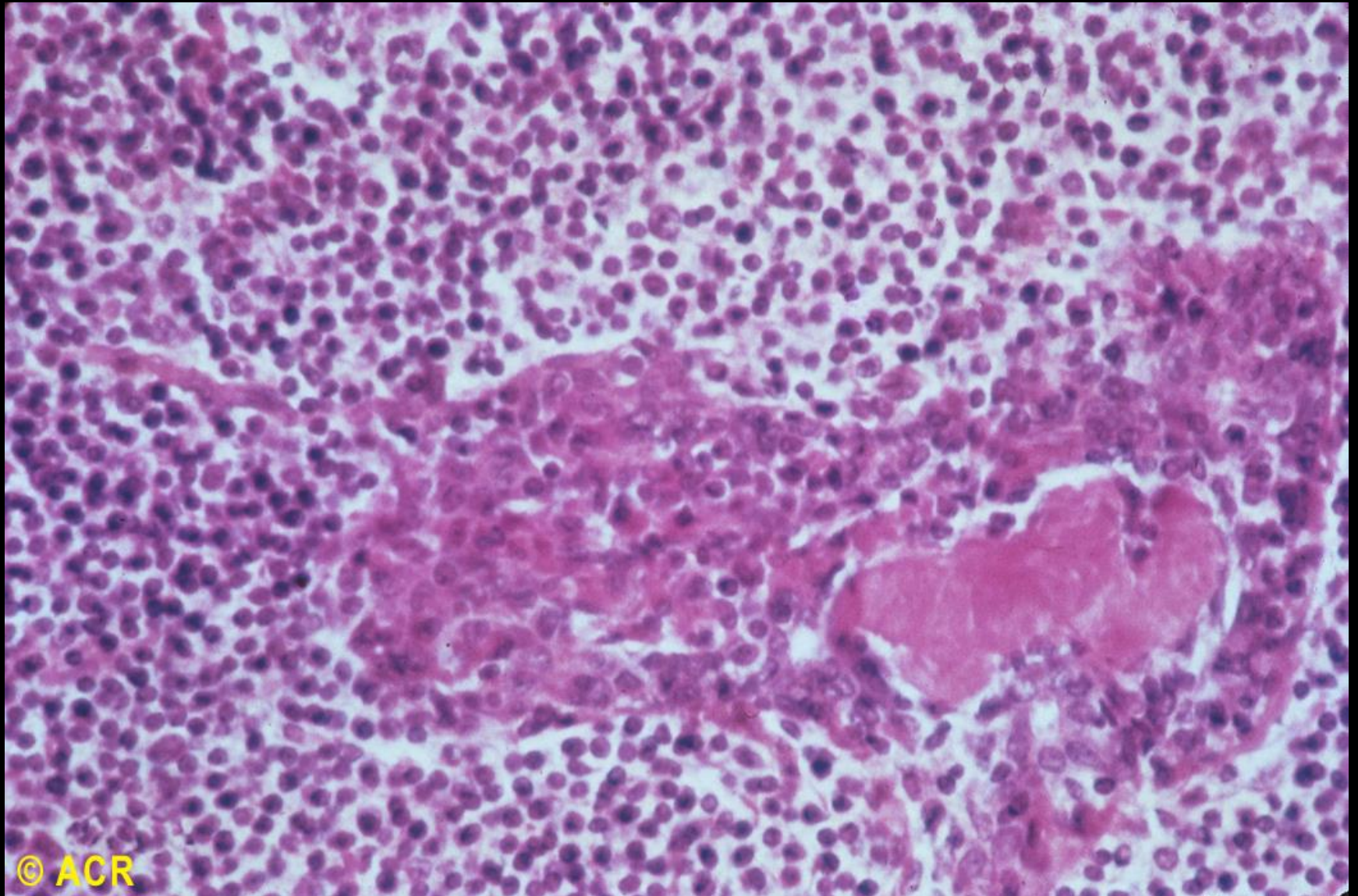
Sjögren's disease: Schirmer test



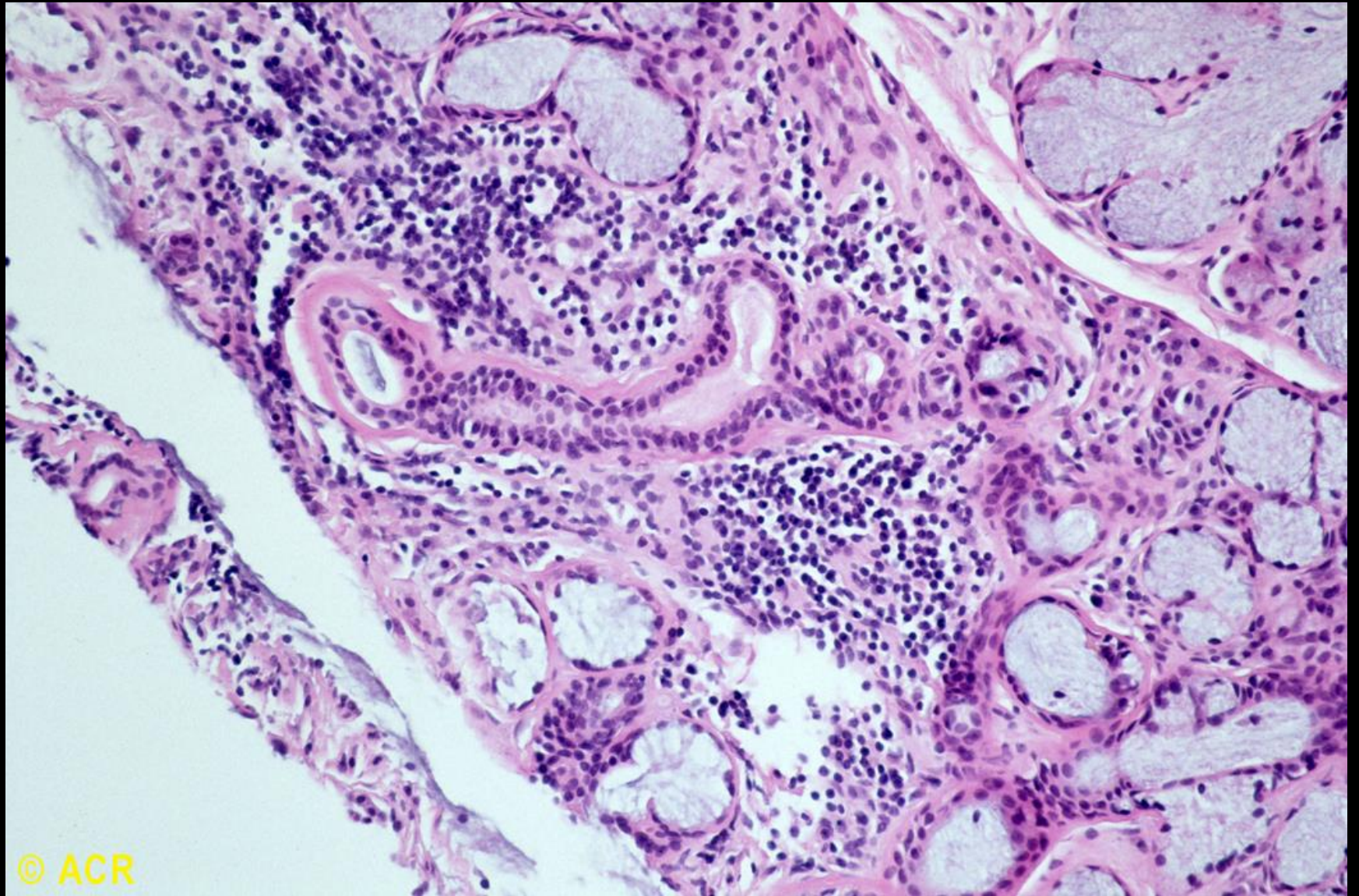
Sjögren's disease: xerostomia



Sjögren's: parotid gland (photomicrograph)



Sjögren's: lip (photomicrograph)



Sjögren's: Treatment

Ocular:

- artificial tears
- ocular cyclosporine
- punctate plugs

Oral:

- appropriate dental care and prevention
- artificial saliva
- muscarinic agonists
 - pilocarpine
 - cevimeline

Systemic:

immunomodulator (hydroxychloroquine, leflunomide, mycophenolate)

B cell depletion (rituximab or belimumab?)(ianalumab, nipocalimab?)

Sjögren's disease: Summary

- Systemic multi-organ autoimmune condition; formerly syndrome
- Commonly manifested as the sicca syndrome
- Multiple associated autoimmune conditions
- Monitor for potential complications of the disease
- Multidisciplinary approach may be needed
 - Rheumatology
 - Nephrology
 - Neurology
 - Ophthalmology
 - Gastroenterology

REVIEW ARTICLE

MECHANISMS OF DISEASE

IgG4-Related Disease

John H. Stone, M.D., M.P.H., Yoh Zen, M.D., Ph.D.,
and Vikram Deshpande, M.D.

IGG4-RELATED DISEASE IS A NEWLY RECOGNIZED FIBROINFLAMMATORY CONDITION characterized by tumefactive lesions, a dense lymphoplasmacytic infiltrate rich in IgG4-positive plasma cells, storiform fibrosis, and, often but not always,

Table 1. Previously Recognized Conditions Now Acknowledged to Fall within the Spectrum of IgG4-Related Disease.

Mikulicz's syndrome (affecting the salivary and lacrimal glands)

Küttner's tumor (affecting the submandibular glands)

Riedel's thyroiditis

Eosinophilic angiocentric fibrosis (affecting the orbits and upper respiratory tract)

Multifocal fibrosclerosis (commonly affecting the orbits, thyroid gland, retroperitoneum, mediastinum, and other tissues and organs)

Inflammatory pseudotumor (affecting the orbits, lungs, kidneys, and other organs)

Mediastinal fibrosis

Retroperitoneal fibrosis (Ormond's disease)

Periaortitis and periarteritis

Inflammatory aortic aneurysm

Idiopathic hypocomplementemic tubulointerstitial nephritis with extensive tubulointerstitial deposits



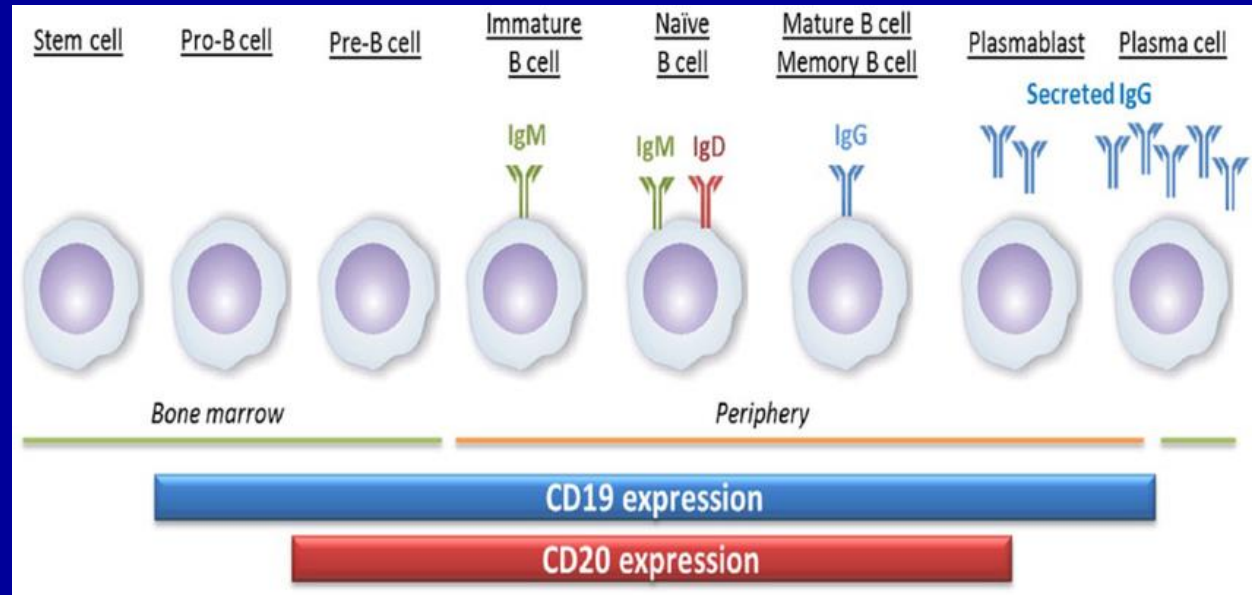
An update on IgG4-related disease

Zachary S. Wallace^a and John H. Stone^b

- Mean age at onset is in the sixth or seventh decade of life
- An elevated serum IgG4 is neither necessary nor sufficient for the diagnosis (since only 1/3 pt's have abnormal)
- Histopathological findings of fibrosis, obliterative phlebitis and mild to moderate tissue eosinophilia form the basis of an IgG4-RD diagnosis.

Management

- glucocorticoids
- azathioprine
- mycophenolate
- rituximab



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ORIGINAL ARTICLE

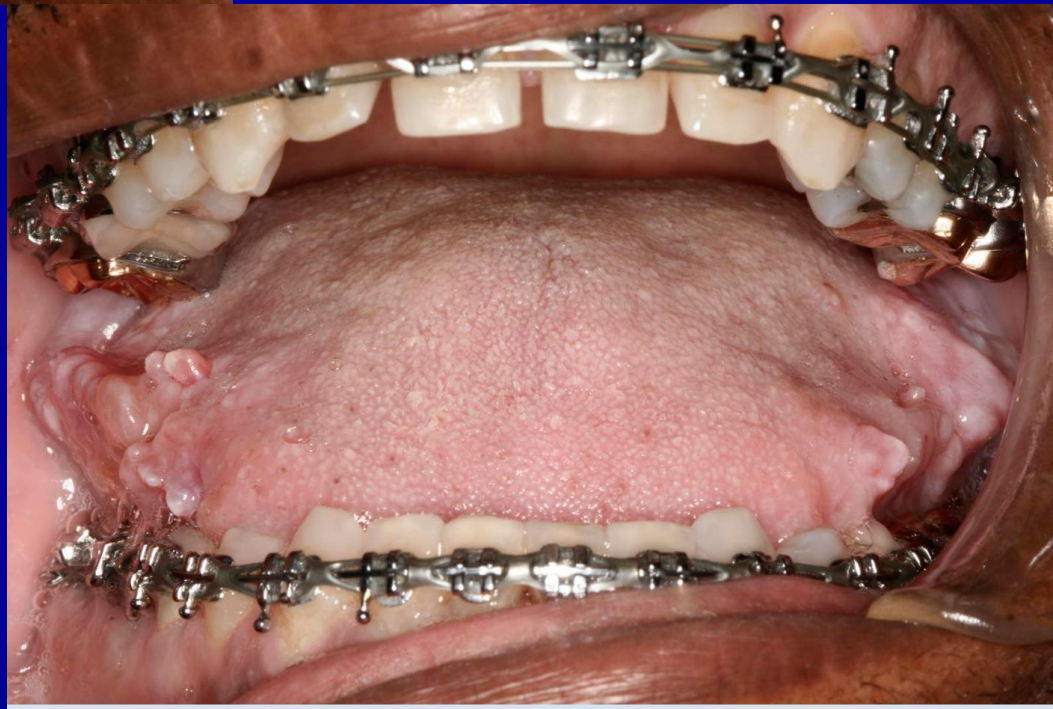
Inebilizumab for Treatment of IgG4-Related Disease

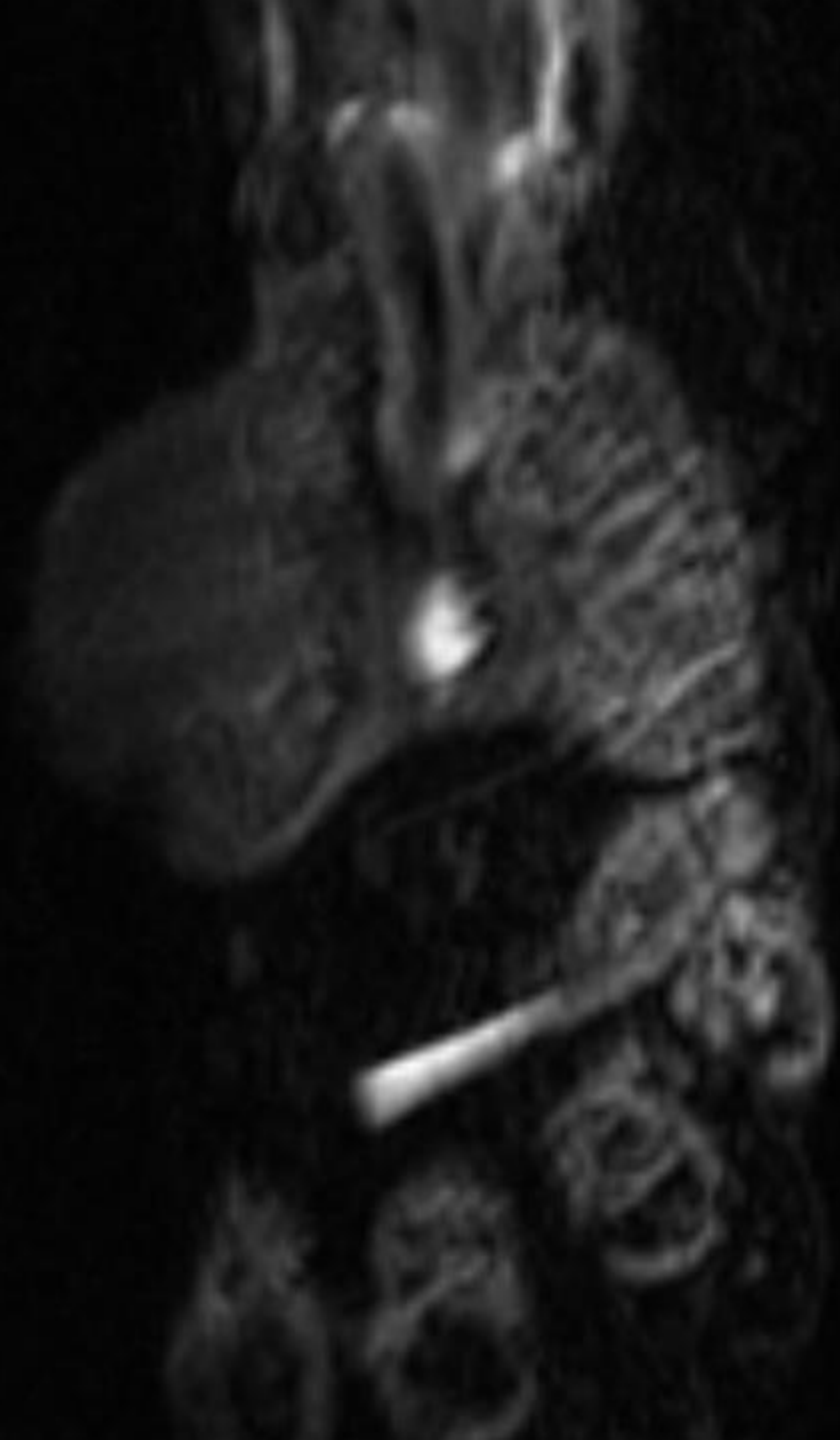
J.H. Stone,¹ A. Khosroshahi,² W. Zhang,³ E. Della Torre,⁴ K. Okazaki,⁵ Y. Tanaka,⁶ J.M. Löhr,⁷ N. Schleinitz,⁸ L. Dong,⁹ H. Umehara,¹⁰ M. Lanzillotta,⁴ Z.S. Wallace,¹ M. Ebbo,⁸ G.J. Webster,¹¹ F. Martinez Valle,¹² M.K. Nayar,¹³ C.A. Perugino,¹ V. Rebours,¹⁴ X. Dong,¹⁵ Y. Wu,¹⁵ Q. Li,¹⁵ N. Rampal,¹⁵ D. Cimbora,¹⁵ and E.L. Culver,¹⁶ for the MITIGATE Trial Investigators*

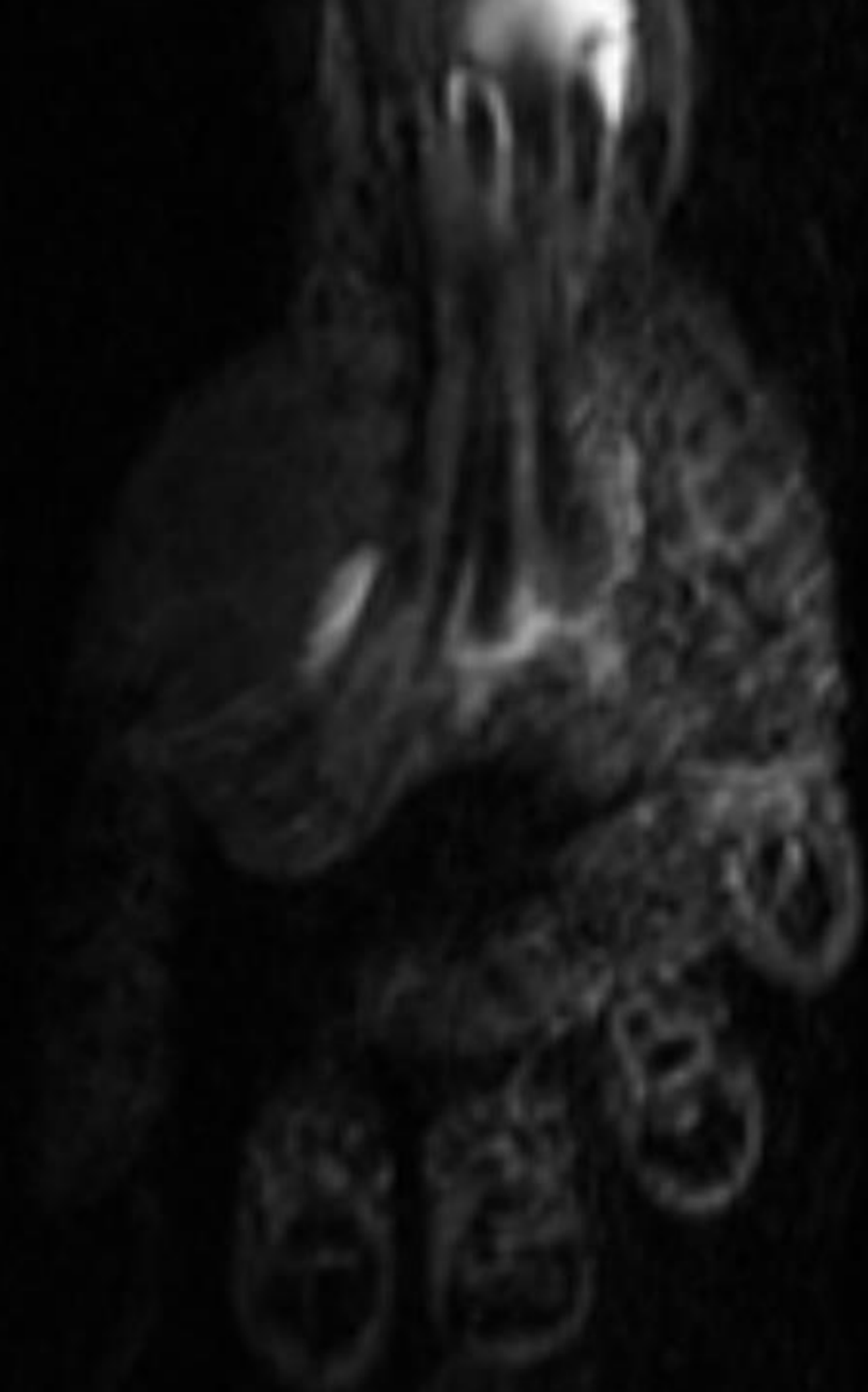
Case 2

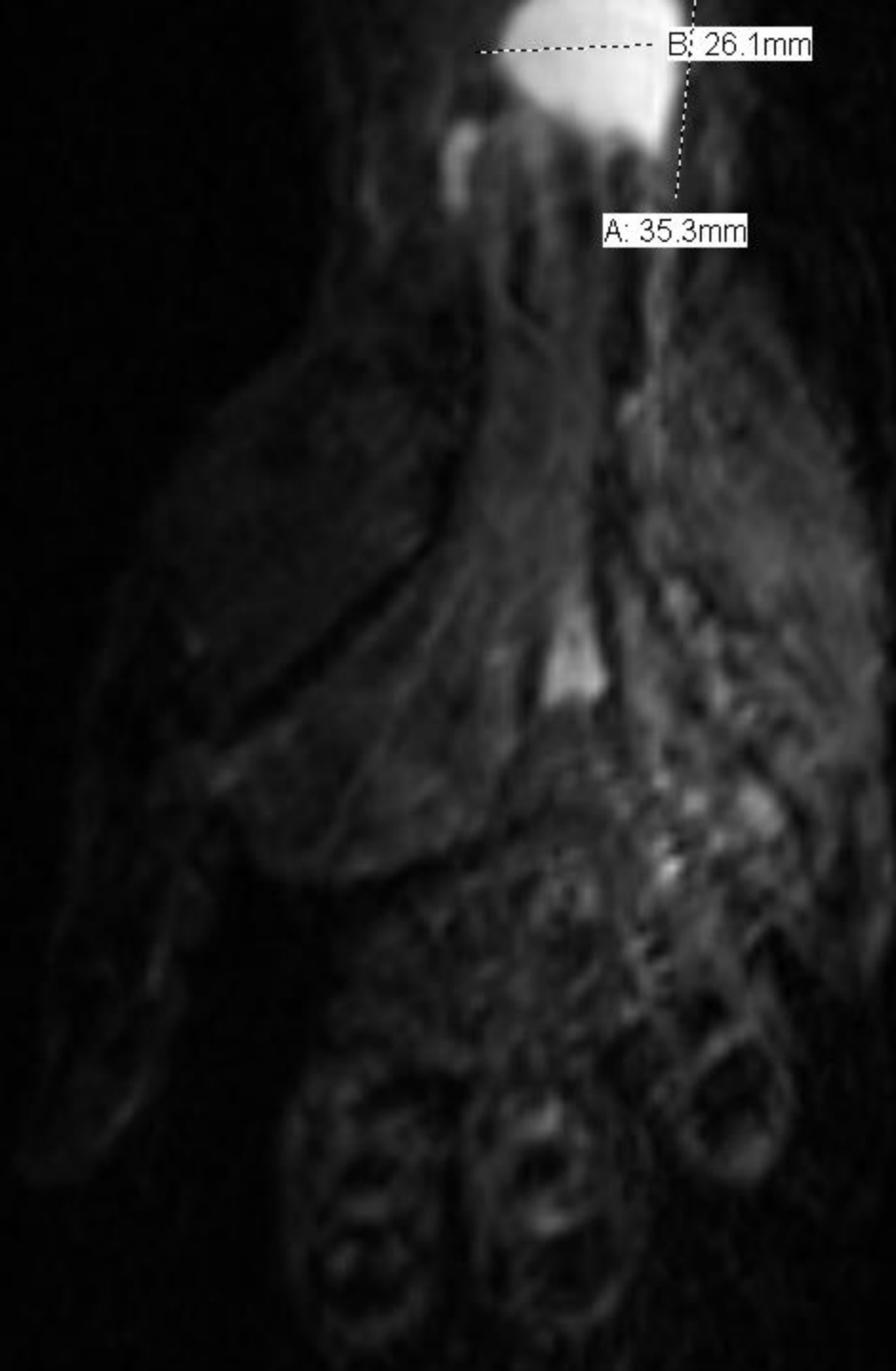
52 year old AA man with 2 months of moderate shoulder aching and stiffness referred from orthopedic clinic following MRI of shoulders that demonstrates synovitis. Also, weeks of shiny irritated nodules on nose and around eyes.

PMH: carpal tunnel, s/p surgery bilaterally; hypertension



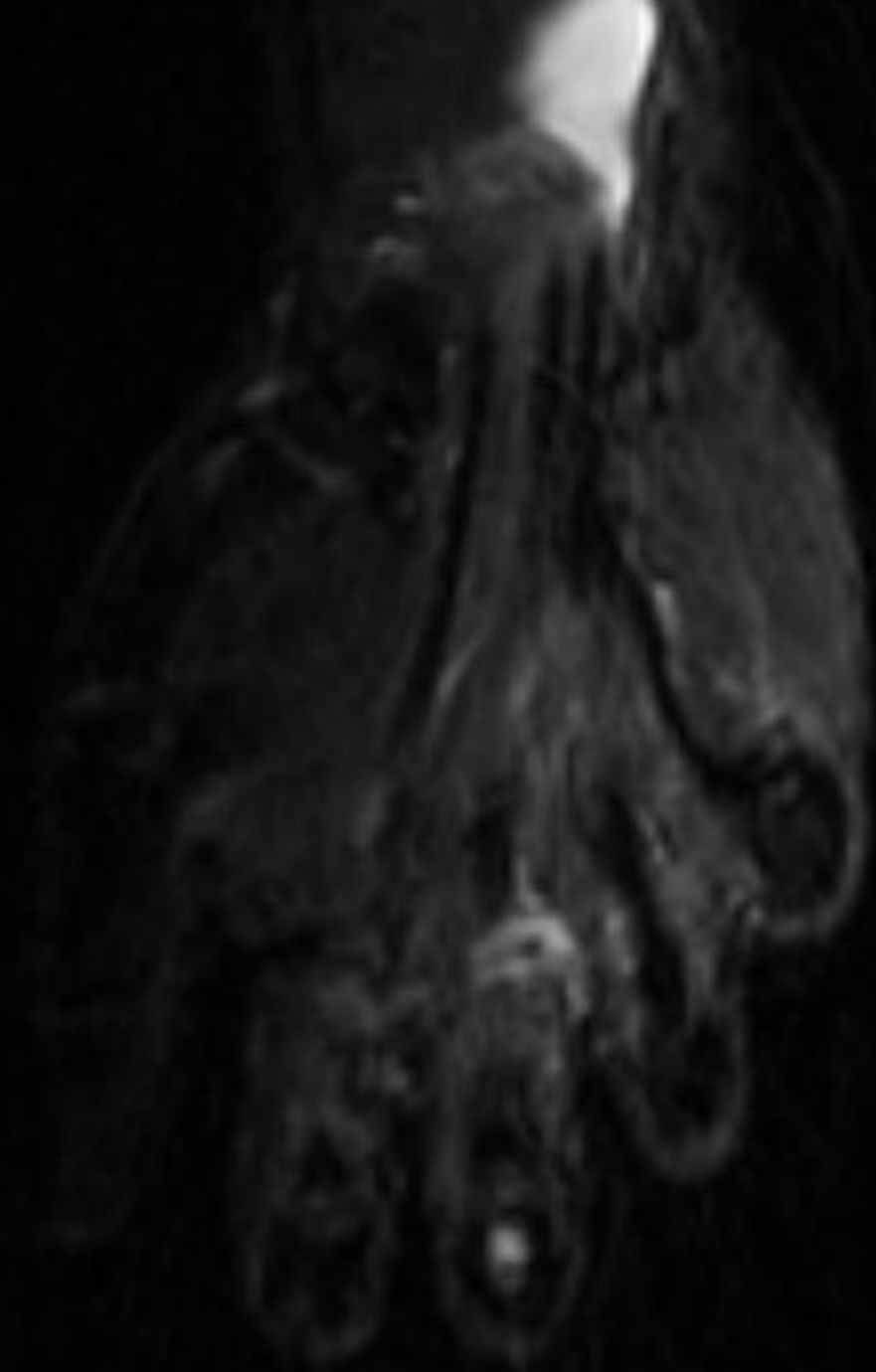


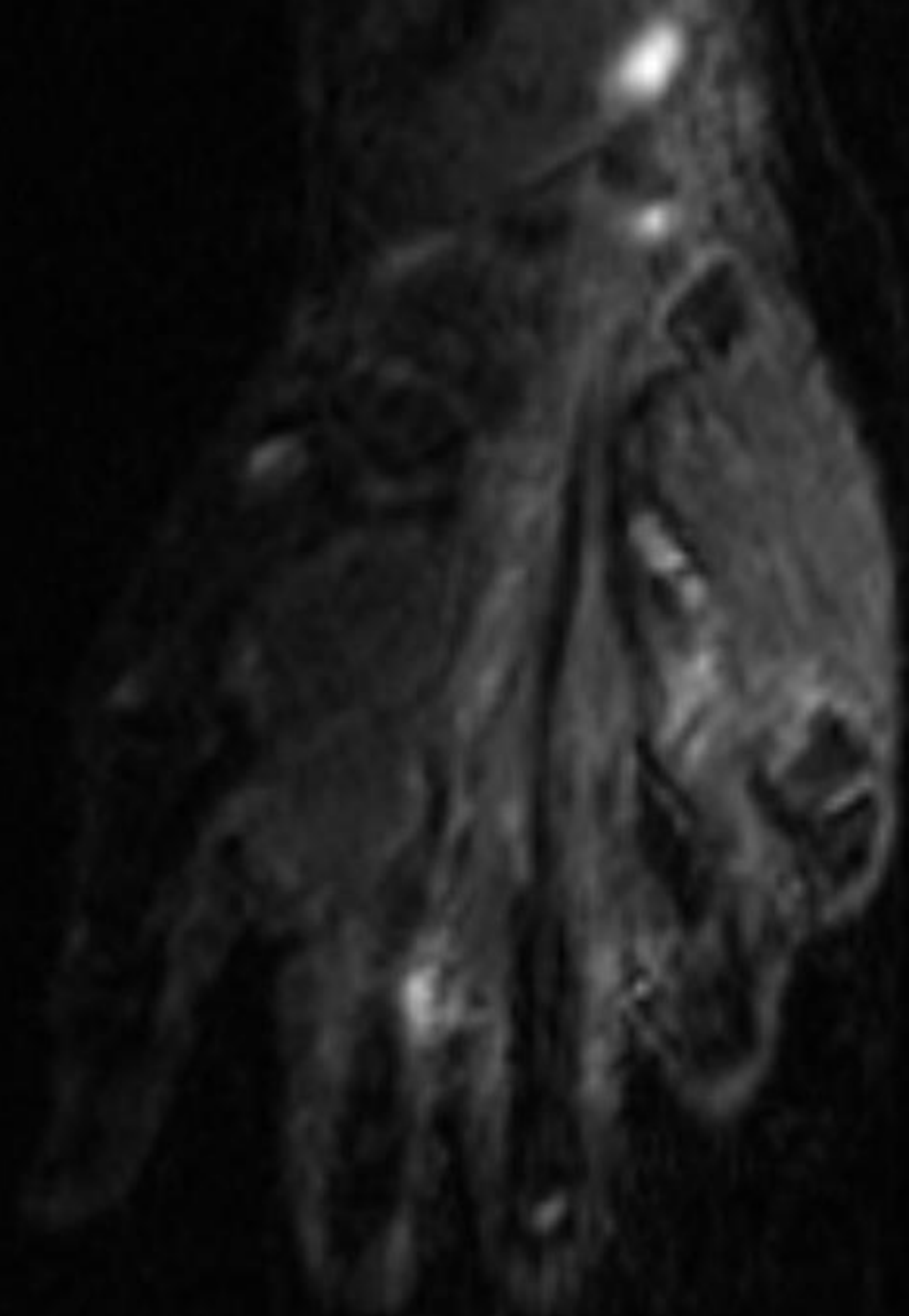




B: 26.1mm

A: 35.3mm





What is the best explanation?

- Sarcoidosis
- Lupus
- Rheumatoid arthritis
- Amyloidosis
- Chronic recurrent multifocal osteomyelitis

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Case 3

- 54 yo native Nigerian man admitted for one month following gradually worsening **fevers**, up to 102F, mostly in **evenings**. Moderate aching in knees, ankles, feet and hands with some **swelling**. **Fatigue** is severe and 20 lbs **wt loss**. Itch on abdomen with mild **redness**.
- No recent travel, no FH.



Component	ESR WEST	ALB	FERRITIN	CRP
Latest Ref R	0 - 15 MM/HR	3.3 - 4.8 G/DL	30 - 400 ng/mL	<10.0 mg/L
2/9/2011 	35 (H)	2.9 (L)	3144.0 (H)	

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2/9/2011 	35 (H)	2.9 (L)	3144.0 (H)	
2/21/2011 	66 (H)		35786.0 (H)	44.0 (H)

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Latest Ref R	0 - 15 MM/HR	3.3 - 4.8 G/DL	30 - 400 ng/mL	<10.0 mg/L
2/9/2011 	35 (H)	2.9 (L)	3144.0 (H)	
2/21/2011 	66 (H)		35786.0 (H)	44.0 (H)
2/23/2011 	97 (H)	3.1 (L)	10762.0 (H)	44.0 (H)

Still's disease

Component	ESR WEST	ALB	FERRITIN	CRP
Latest Ref R	0 - 15 MM/HR	3.3 - 4.8 G/DL	30 - 400 ng/mL	<10.0 mg/L
2/9/2011 	35 (H)	2.9 (L)	3144.0 (H)	
2/21/2011 	66 (H)		35786.0 (H)	44.0 (H)
2/23/2011 	97 (H)	3.1 (L)	10762.0 (H)	44.0 (H)
3/8/2011 	34 (H)	3.7	2528.0 (H)	7.0
3/30/2011 	13		1094.0 (H)	<5.0
4/28/2011 	34 (H)		465.2 (H)	7.0
5/26/2011 	22 (H)		215.3	

Comparison Chart of Systemic Autoinflammatory Diseases (SAID)

[illegible]

Questions?

